

A Mean-Reversion Model for US Sectors

Exploiting Price-Reversal Opportunities Through Liquid ETFs



CONCRETUM RESEARCH

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Last week, inspired by the sector rotation we have recently observed across financial markets, we introduced a **short-term mean-reversion framework applied to three industry ETFs**.



Profiting From Sector Dispersion

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The positive response from readers naturally led us to a broader question:

Can the same idea be extended into a diversified portfolio of sector ETFs?

In this article, we apply the exact same framework to the complete universe of eleven SPDR Sector ETFs, creating an actively managed portfolio that dynamically allocates capital only when statistically significant reversal opportunities emerge.



SPDR SECTOR ETFs

OUR UNIVERSE OF 11 SECTOR ETFs

TICKER	SECTOR NAME	TICKER	SECTOR NAME
 XLB	Materials	 XLU	Utilities
 XLE	Energy	 XLV	Health Care
 XLF	Financials	 XLY	Consumer Discretionary
 XLI	Industrials	 XLRE	Real Estate
 XLK	Technology	 XLC	Communication Services
 XLP	Consumer Staples		



Inception dates: All ETFs listed on 22 Dec 1998, except XLRE (08 Oct 2015) and XLC (19 Jun 2018).

Source: State Street Global Advisors (SSGA) | Data as of July 2026

The results are encouraging.

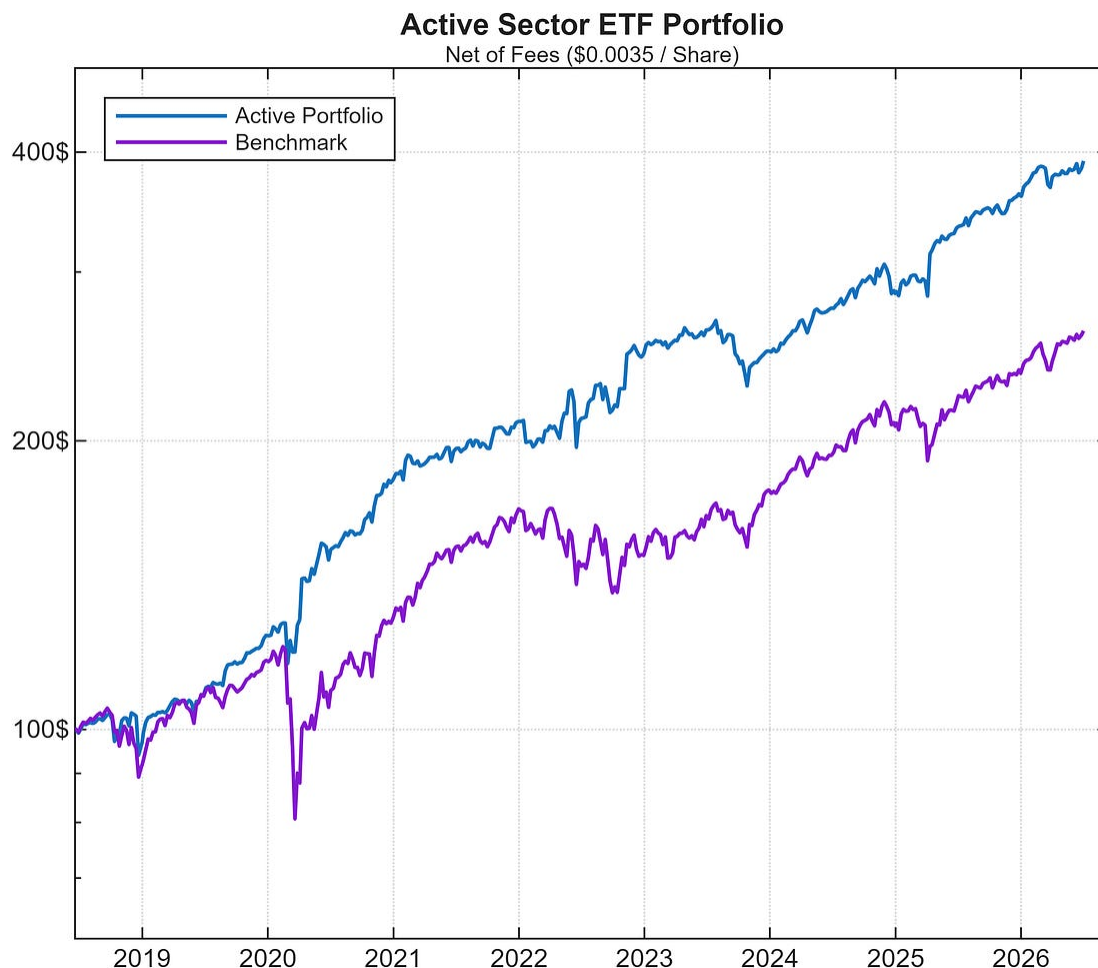
Over the period from June 2018 through early July 2026, the strategy compounds at **18.5% per year after transaction costs**, comfortably outperforming a passive equal-weight benchmark, which returns **12.6% annually** over the same period.

At the same time, it improves the portfolio's **Sharpe ratio from 0.75 to 1.33**, reduces the **maximum drawdown from -36.3% to -14.7%**, and achieves these results while **deploying only 58% of its capital on average**, leaving a substantial portion unallocated throughout much of the sample.

For simplicity, we do not account for the interest that this idle cash could have earned, meaning the reported returns are likely conservative.

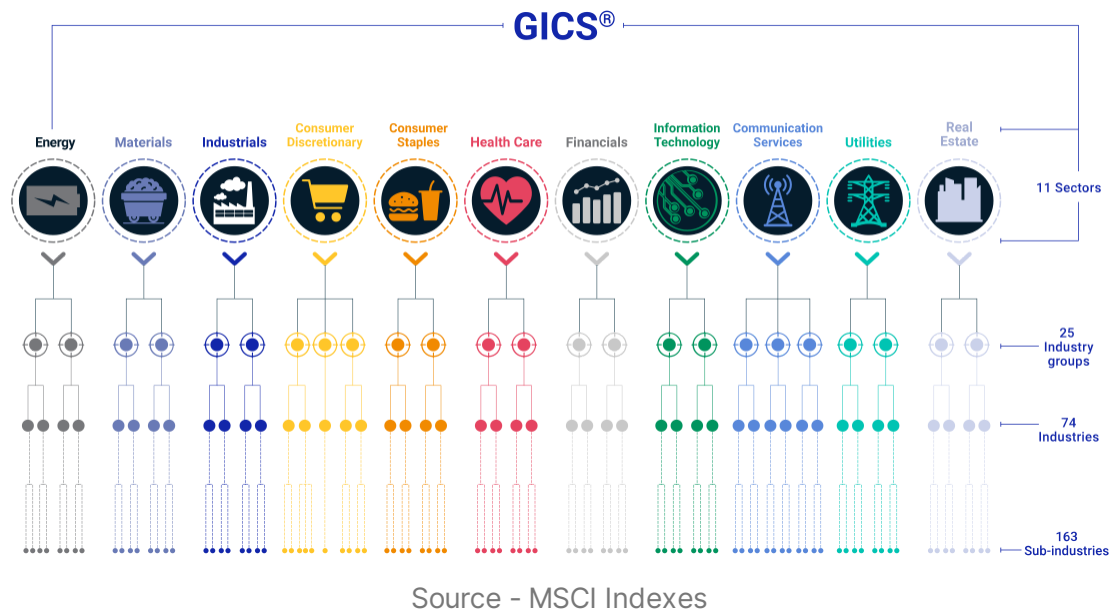
In the remainder of this article, we unpack the full methodology behind the strategy, examine the empirical behavior supporting the signal, and

show how the individual trades are combined into a diversified portfolio. We then evaluate its performance after transaction costs, capital usage, and resilience across different market environments.



Universe Construction

Whenever we take on work that touches broad sectors or industry exposures, our reference classification system is **GICS®**, the Global Industry Classification Standard developed jointly by MSCI and Standard & Poor's in 1999, and now maintained by MSCI and S&P Dow Jones Indices.



It consists of a four-tier hierarchy that sorts every listed company first into one of eleven sectors, then into progressively finer industry groups, industries, and sub-industries.

As most traders will know, **State Street** provides a series of eleven liquid ETFs designed to track the performance of official GICS® sectors. These funds constitute our investable universe, and the following table presents them along with their listing dates.

Ticker	Name	Inception Date
XLB	State Street Materials Select Sector SPDR ETF	22 Dec 1998
XLE	State Street Energy Select Sector SPDR ETF	22 Dec 1998
XLF	State Street Financial Select Sector SPDR ETF	22 Dec 1998
XLI	State Street Industrial Select Sector SPDR ETF	22 Dec 1998
XLK	State Street Technology Select Sector SPDR ETF	22 Dec 1998
XLP	State Street Consumer Staples Select Sector SPDR ETF	22 Dec 1998
XLU	State Street Utilities Select Sector SPDR ETF	22 Dec 1998
XLV	State Street Health Care Select Sector SPDR ETF	22 Dec 1998
XLY	State Street Consumer Discretionary Select Sector SPDR ETF	22 Dec 1998
XLRE	State Street Real Estate Select Sector SPDR ETF	08 Oct 2015
XLC	State Street Communication Services Select Sector SPDR ETF	19 Jun 2018

Given that the **SPDR Communication Services ETF** (the last of the eleven to come to market) was only added on June 19, 2018, our historical sample spans the period from mid-2018 through early-July 2026, the window over which **the basket of sectors is entirely investable**.

The data used throughout this study is fetched from Massive (formerly Polygon.io) both for end-of-day prints and for the **intraday snapshots**

we will rely on later.

Signal Generation

Our entry-exit signals follow the same criteria outlined in our original piece, which we **repeat here for convenience**.

We define the core buy and sell conditions on symmetrical IBS readings, with entries triggered when

$$\frac{\text{Close}_{15:45} - \text{Low}_{15:45}}{\text{High}_{15:45} - \text{Low}_{15:45}} < 0.1$$

and exits triggered when

$$\frac{\text{Close}_{15:45} - \text{Low}_{15:45}}{\text{High}_{15:45} - \text{Low}_{15:45}} > 0.9$$

Since IBS is meant to capture directional conviction, we find it **economically sound** to condition our entries and exits on sessions where the price move itself was somewhat meaningful, rather than just expected “noise.”

To do so, for every ETF, we compare each day’s realized range through 15:45 with that same ETF’s own recent range history. We define the daily (log) range as

$$\text{Range}_t = \ln\left(\frac{\text{High}_t}{\text{Low}_t}\right)$$

and scale today’s value (measured through 15:45) by the volatility of the prior 21 sessions’ full-day (log) ranges, lagged by one day so no information about today’s session leaks into the scale itself:

$$\text{Relative Range}_t = \frac{\ln(\text{High}_{15:45,t} / \text{Low}_{15:45,t})}{\text{StdDev}(\text{Range}_{t-1}, \dots, \text{Range}_{t-21})}$$

Importantly, we only act on an IBS signal when this relative range value exceeds 1, i.e. only on days whose range through 15:45 already runs **above 1σ** , signaling higher-than-average range expansion.

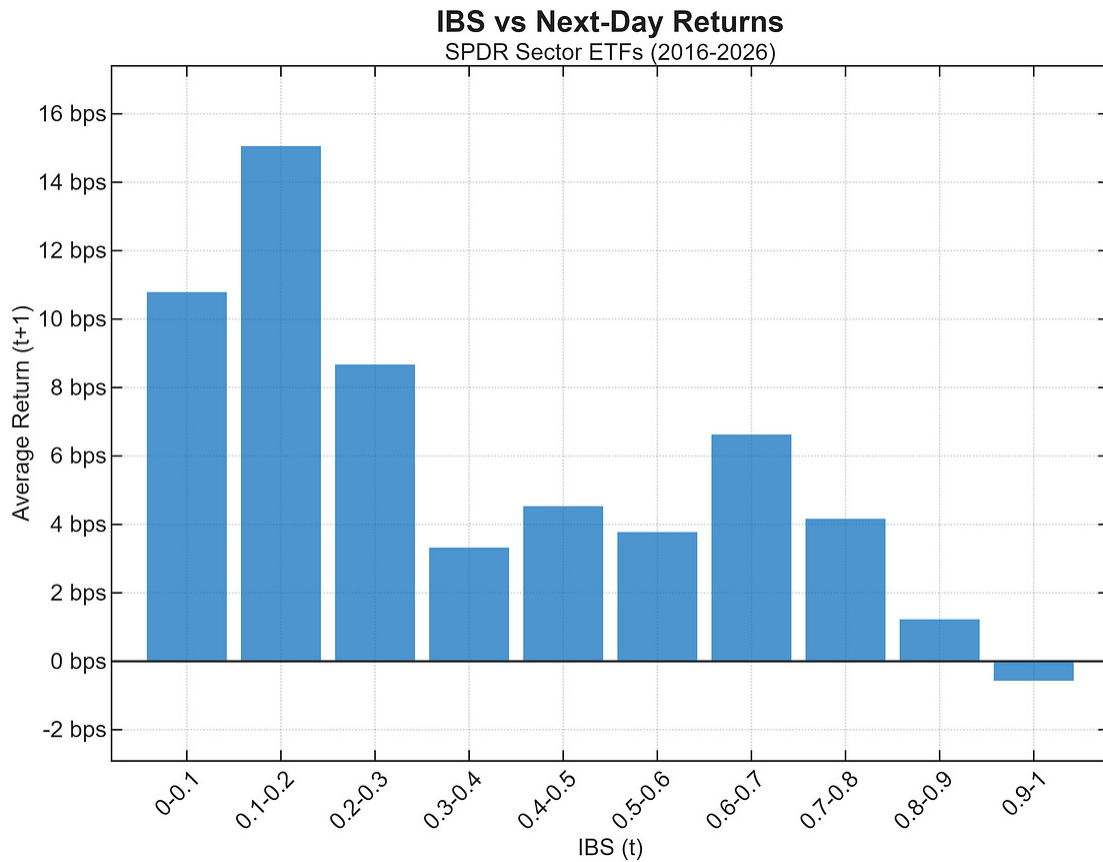
Note - The intraday OHLC snapshot is captured 15 minutes before market close: 15:45 on normal trading days, or 12:45 on early-close days. If no price update is available at the required timestamp, the most recent prior update is used instead.

Additional IBS Tests

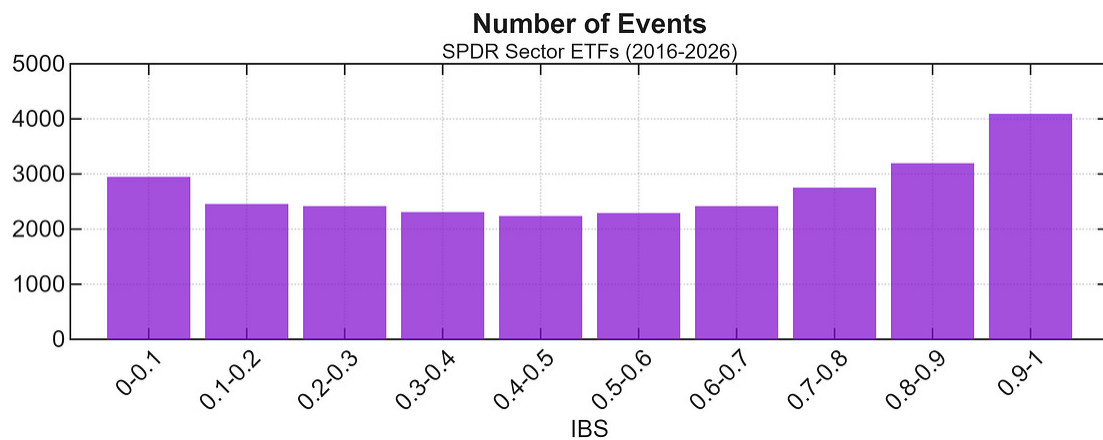
Given that **Internal Bar Strength (IBS) remains a core component of our signal**, we are curious to test whether the relationships documented in the academic literature also hold true in our own dataset.

As readers may recall, in *The IBS Effect: Mean Reversion in Equity ETFs* (2013), Pagonidis suggests that low IBS readings tend to precede above-average next-day returns, while high IBS readings are typically followed by below-average returns.

To assess whether this effect also holds in our universe, we analyze SPDR Sector ETF data over the past decade and perform a simple test: we record each ETF's daily IBS reading and **compute the average next-day return across ten IBS buckets**, ranging from 0 to 1 in increments of 0.1.



What we observe, consistent with what Pagonidis found on **data through 2013**, is a tendency for **high IBS values to precede poor day-ahead returns** (marginally negative in our 0.9-1 bucket), whereas **low IBS values appear to precede more positive subsequent returns**, albeit with some noise.



Interestingly, if we count the number of events within each bucket, we notice a **U-shaped pattern**, with the more populated buckets concentrated at extreme IBS readings (below 0.1 and above 0.7-0.8). As also highlighted by Pagonidis (2013), *the shape of the distribution is an expected property of the random price process*.

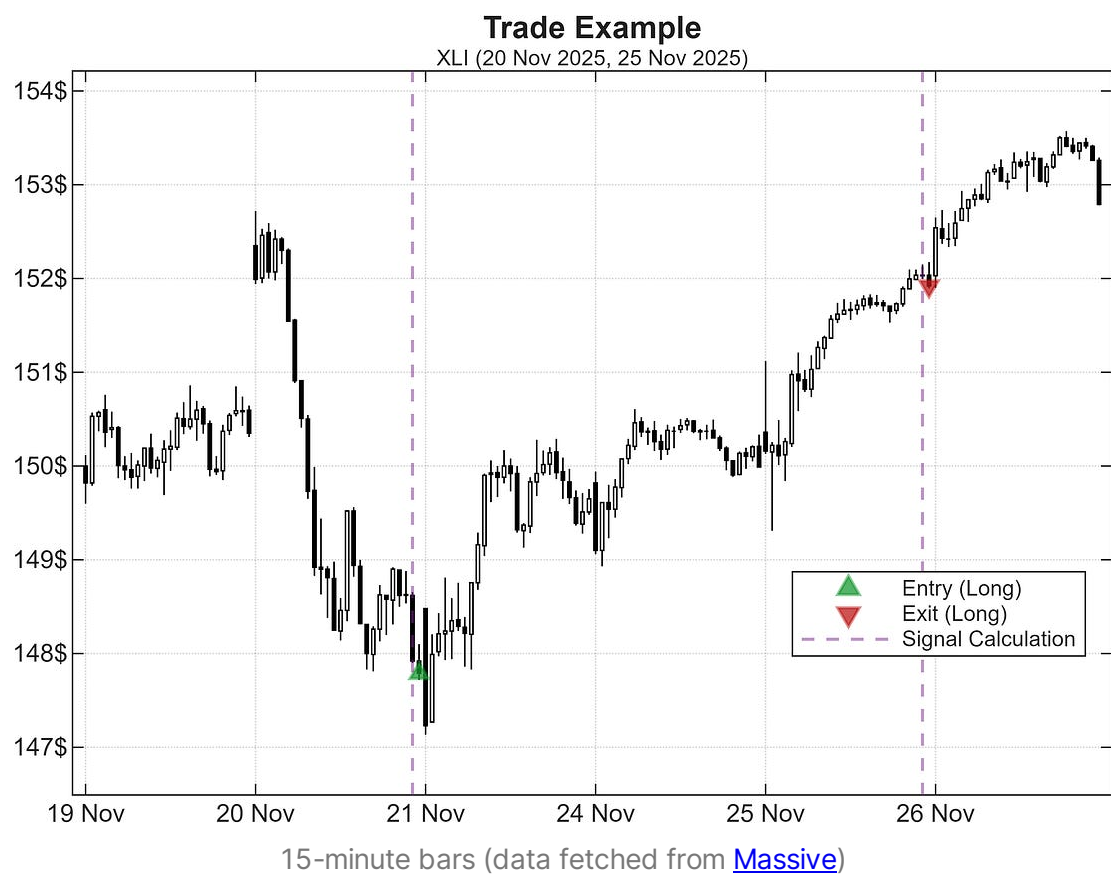
Results

For continuity with the original analysis, we report the formula behind our position-sizing policy:

$$\text{Shares}_{i,t} = \frac{1.5 \times \text{AUM}_{t-1}}{N \times P_{i,t}^{15:45}}$$

where N represents the size of our investable universe and $P_{i,t}$ is the 15:45 snapshot price at which ETF i is purchased at the close of day t .

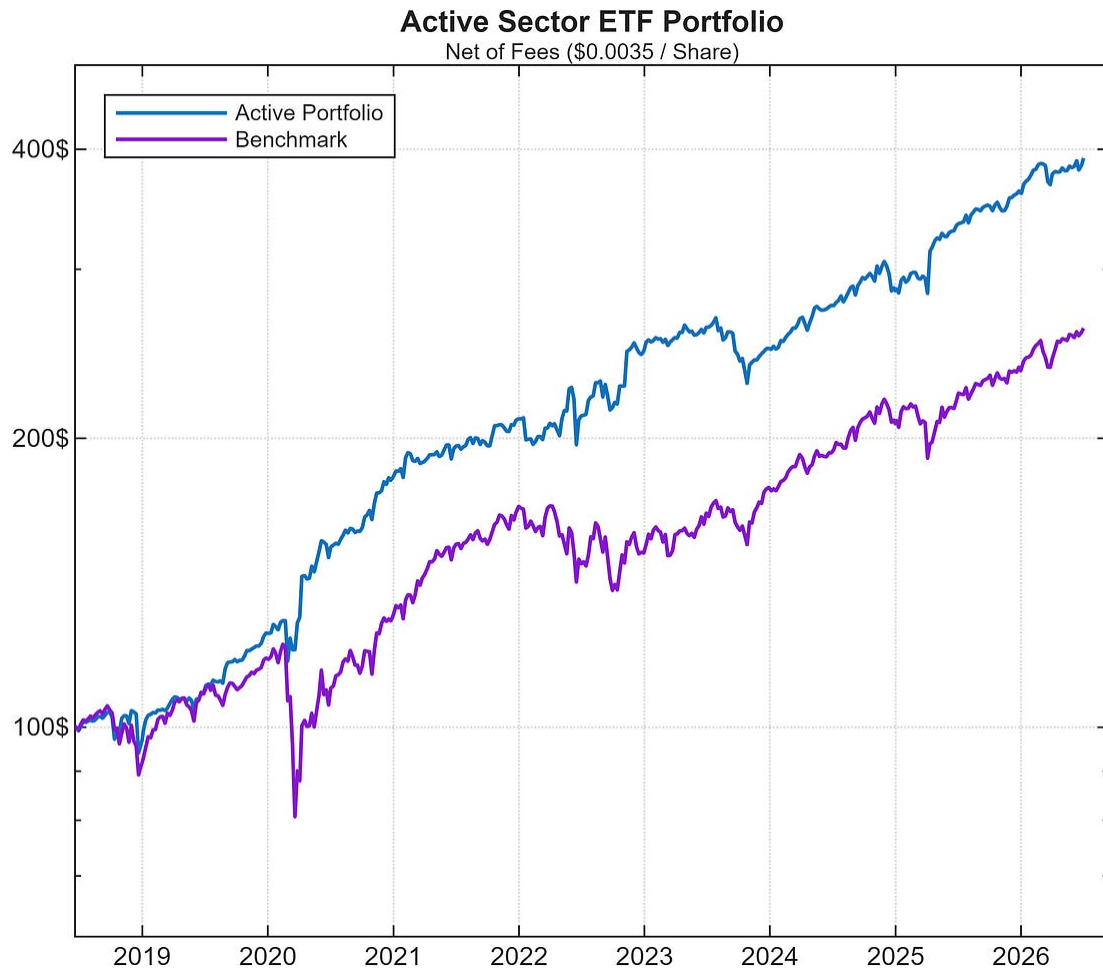
Below, we show an **illustrative trade example on XLI** (the Industrial Select Sector ETF), with entry on November 20, 2025, and exit on November 25, 2025. The figure highlights the 15:45 snapshot used to compute the signal on both sides of the trade, followed by the market-on-close (MOC) fill fifteen minutes later.



The table that follows summarizes the remaining assumptions used in our historical simulation.

Assumption	Explanation
Returns	Returns include reinvestment of dividends and are adjusted for splits
Position Management	Once initiated, positions are never touched until liquidation
Fees	Standard IBKR tier of \$0.0035 / Share, doubled on sell transactions to conservatively account for additional sell-side charges (no hard minimums are applied)
Fractional Shares Trading	Trading of fractional shares is allowed
Interest on Cash	No interest earned on idle cash balance
Cost of Leverage	No financing cost is charged on leverage
Benchmark	Synthetic Index that holds all 11 SPDR Sector ETFs (positions are equally-weighted and rebalanced daily)

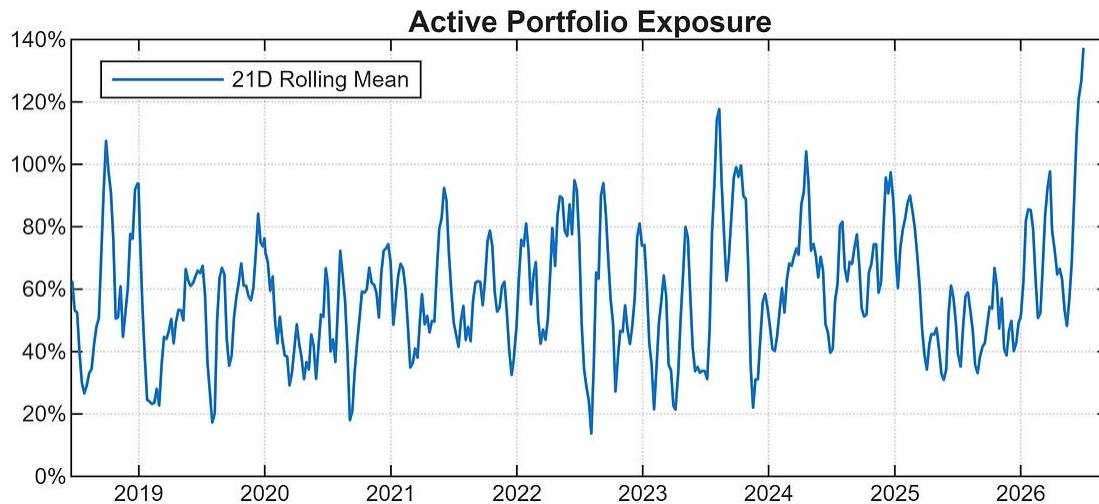
We notice that, across our sample, the **net-of-fee outperformance of the actively traded book against its benchmark is rather clear.**



As summarized in the following table, the active book compounds at **18.5% per year against the benchmark's 12.6%**, while cutting maximum drawdown by more than half. Additionally, our mean-reversion framework exhibits a Sharpe ratio of ~1.33, representing a meaningful improvement over the 0.75 reported for our synthetic benchmark.

Strategy	CAGR	Sharpe (0% Rf)	Max Drawdown	Avg Exposure
Benchmark	12.6%	0.75	-36.3%	100.0%
Active Portfolio	18.5%	1.33	-14.7%	58.4%

Importantly, this net-of-fee performance was achieved with an **average portfolio exposure of around 58%** and only occasional stretches above 100% (along with a maximum recorded value of ~157%), as highlighted in the figure below.



In practice, the cash the portfolio holds for much of the sample would earn interest that, for simplicity, we have chosen not to consider, **providing an additional positive drift** on top of the results shown.

Conclusion

From what we have seen, pairing IBS with a straightforward range-expansion overlay has generated **compelling net-of-fee results**, and the exercise reinforces the value of IBS as a building block for reversal-type strategies.

Notably, these net-of-fee results **beat a passive benchmark allocation even in recent years**, a period in which different mean-reversion approaches have struggled to deliver attractive performance, with much of their historical edge having faded through the 1990s and 2000s (see Da, Liu, and Schaumburg (2014) for additional details).

Looking ahead, we believe this portfolio **still has room to grow**, whether by **widening the universe** to other industry ETFs and asset classes, or by adopting **more nuanced position-sizing** rules than the simple equal-weight scheme used here.

As always, we hope this piece leaves our readers with a **few ideas worth building on for their research and trading practice**.

If you found this article useful, feel free to leave a comment and reach out via direct message or email at info@concretumgroup.com for any questions.



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